URECA - Three Languages of Quantum Mechanics: a Mathematical Tour

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1 Mathematical Foundation

Definition 1.1 (Hilbert Space) *A Hilbert space is a complex vector space $\mathcal{H}$, endowed with an Hermitian inner product $\langle\cdot|\cdot\rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$ such that its associated topology is Cauchy complete.*

(A) We recall that an Hermitian inner product satisfies the following properties:
1) Conjugate symmetry:

$$\langle x|y\rangle = \overline{\langle y|x\rangle}, \quad \text{for all } x, y \in \mathcal{H}$$

2) Linearity on the second argument:

$$\langle x|ay_1 + by_2\rangle = a\langle x|y_1\rangle + b\langle x|y_2\rangle, \quad \text{for all } x, y_1, y_2 \in \mathcal{H}, \text{ for all } a, b \in \mathbb{C}$$

3) Positive definiteness:

$$\langle x|x\rangle \geq 0, \quad \text{for all } x \in \mathcal{H} \text{ and } = 0 \text{ iff } x = 0.$$

(B) We recall that Cauchy complete means that all sequences in $\mathcal{H}$ that are Cauchy with respect to the norm induced by the inner product $\langle\cdot|\cdot\rangle$ must converge in $\mathcal{H}$ (see later in the Topology course).

The Hilbert space is the mathematical space where we describe all quantum mechanic states. All the states can be added, rotate and projected like vectors, and the inner products describe the corresponding probability of getting the corresponding result. Everything we do in quantum mechanics are just geometric relation between vectors and operators in the Hilbert space. Therefore, the Hilbert space is the theoretical foundation of quantum mechanics.

Remark 1.2 (About completeness) *All finite dimensional vector spaces are automatically Cauchy complete (hence Hilbert spaces), but this is not true for infinite dimensional ones and this must be imposed. For instance $L^2(\mathbb{R}^3, \mathbb{C})$ (square integrable functions) endowed with the L^2 Hermitian product is a Hilbert space while $C^\infty(\mathbb{R}^3, \mathbb{C})$ (smooth functions) is not. We will not consider this property in this project.*

Definition 1.3 (Operator) *An operator on a Hilbert space is a complex linear map $A : \mathcal{H} \to \mathcal{H}$. If $\mathcal{H}$ is finite dimensional, we can identify A with its matrix with respect to a basis.*

Remark 1.4 (About boundedness) *All operators on finite dimensional Hilbert spaces are automatically continuous (also called bounded). This is not true in general and one imposes the continuity in the definition. We will not consider this in this project.*

Definition 1.5 (Adjoint of an Operator) *The adjoint of an operator $A : \mathcal{H} \to \mathcal{H}$ is the operator $A^\dagger : \mathcal{H} \to \mathcal{H}$ defined by*

$$\langle A^\dagger x | y \rangle = \langle y | Ax \rangle, \quad \forall x, y \in \mathcal{H}.$$

The matrix of the adjoint operator is transpose conjugate of the matrix of the operator.

Definition 1.6 (Unitary Operator) *An operator $U : \mathcal{H} \to \mathcal{H}$ is unitary if its inverse is equal to its conjugate transpose:*

$$UU^\dagger = U^\dagger U = I.$$

Definition 1.7 (Hermitian Operator) *An operator $H : \mathcal{H} \to \mathcal{H}$ is Hermitian if it is equal to its conjugate transpose:*

$$H = H^\dagger.$$

Definition 1.8 (Skew Hermitian Operator) *An operator $A : \mathcal{H} \to \mathcal{H}$ is skew-Hermitian if it is equal to the negative of its conjugate transpose:*

$$-A = A^\dagger.$$

Definition 1.9 (Projection Operator) *An operator $P : \mathcal{H} \to \mathcal{H}$ is a projection if its square is equal to itself:*

$$P^2 = P.$$

Furthermore, if it is orthogonal if it is equal to its conjugate transpose:

$$P = P^\dagger.$$

Remark 1.10 (Properties of Hermitian, skew Hermitian and unitary matrices)

1. *All diagonal entries are real: $a_{ii} \in \mathbb{R}$*
2. *Eigenvectors respective to different eigenvalues are orthogonal:*

$$A\mathbf{v_1} = \lambda_1 \mathbf{v_1},\ A\mathbf{v_2} = \lambda_2 \mathbf{v_2} \wedge \lambda_1 \neq \lambda_2 \implies \langle \mathbf{v_1} | \mathbf{v_2} \rangle = 0.$$

Theorem 1.11 (Singular Value Decomposition) *It is used to convert one matrix into the product of a unitary matrix, a diagonal matrix and another unitary matrix. i.e., every linear transformation is a combination of a rotation (into the direction of the eigenvectors), a stretching (along the eigenvectors), and another rotation:*

$$A = U\Sigma V^T$$

where Σ is rectangularly diagonal and has its entries as the singular values of A in descending order; U and V are orthogonal and they are composed of the left and right singular vectors of A (normalized eigenvectors of AA^T and A^TA).

Definition 1.12 (Bra–Ket Notation) *In quantum mechanics, a physical state is represented by a vector in a Hilbert space. Dirac introduced the bra–ket notation as a compact and elegant way to describe such vectors and their inner products.*

Ket vector. *A ket is a normalized column vector of the form*

$$|\psi\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix},$$

where $a_i \in \mathbb{C}$.

Bra vector. *The Hermitian conjugate of a ket vector is called a bra vector:*

$$\langle\psi| = \begin{pmatrix} a_1^* & a_2^* & \cdots & a_n^* \end{pmatrix}.$$

Inner product. *The inner product between a bra and a ket is written as*

$$\langle\psi|\psi|\psi|\psi\rangle = \langle\psi|\psi\rangle = \sum_{i=1}^{n} a_i^* a_i = \sum_{i=1}^{n} |a_i|^2.$$

For a normalized state,

$$\langle\psi|\psi|\psi|\psi\rangle = 1.$$

Expectation value of an observable. *Let* $\hat{O}$ *be an observable and let* $\{|a_i\rangle\}$ *be an orthonormal basis. Expanding the state as*

$$|\psi\rangle = \sum_i c_i \, |a_i\rangle \,,$$

where

$$c_i = \langle a_i|\psi|a_i|\psi\rangle \,,$$

the matrix elements of $\hat{O}$ *are defined by*

$$O_{ij} = \langle a_i|\hat{O}|a_j\rangle \,.$$

Then,

$$\hat{O}\,|\psi\rangle = \sum_i \left(\sum_j O_{ij} c_j \right) |a_i\rangle \,,$$

and the expectation value of $\hat{O}$ *is*

$$\left\langle \hat{O} \right\rangle = \langle\psi|\hat{O}|\psi\rangle = \sum_{i,j} c_i^* O_{ij} c_j.$$

If $\{|a_i\rangle\}$ *is an eigenbasis of* $\hat{O}$*, then*

$$O_{ij} = o_i \delta_{ij},$$

and the expectation value simplifies to

$$\langle\psi|\hat{O}|\psi\rangle = \sum_i o_i |c_i|^2.$$

2 Axioms of nonrelativistic quantum mechanics and physical interpretation

Definition 2.1 (Hilbert Space) *A Hilbert space is a complete inner product vector space over the field of complex numbers. In quantum mechanics, it serves as the mathematical space in which quantum states are defined.*

Definition 2.2 (Quantum State) *It is represented by a state vector (or wave function) in Hilbert space. Typically denoted as $|\psi\rangle$.*

In classical mechanics, the state of a system with N degrees of freedom is described by a point in phase space, specified by $2N$ coordinates:

$$(q_1, \ldots, q_N, p_1, \ldots, p_N),$$

where q_i and p_i denote the generalized positions and momenta.

In quantum mechanics, however, the state of a physical system is represented by a vector

$$|\psi\rangle$$

in a Hilbert space $\mathcal{H}$. Physical states are required to be normalized:

$$\langle\psi|\psi\rangle = 1.$$

Observable physical quantities are represented by Hermitian operators acting on the Hilbert space.

Definition 2.3 (Observables and Operators) *In quantum mechanics, the state (observable, i.e. some parameter that can be observed) of an object is described by vectors in Hilbert space, thus we cannot directly observe its state. In order to observe the state, we introduce operator, mathematically a Hermitian matrix, to get the expectation the state we want. For example, we now have a particle in state $|\psi\rangle$. If we want to know the averaged momentum of it, we compute the expectation value of the momentum operator:*

$$\bar{P} = \langle\psi|\hat{p}|\psi\rangle.$$

The quantity $\bar{P}$ is a real number representing the mean momentum of the particle in the state $|\psi\rangle$.

Definition 2.4 (Eigenstates and Eigenvalues) *Mathematically, eigenstates and eigenvalues are just the eigenvectors and corresponding eigenvalues of the Hermitian matrix operator. Physically, the eigenstates denote the possible states that the object could be in after the measurement and the eigenvalues are the results you will get after applying the operator.*

Postulate 2.5 (Measurement Postulate) *When a quantum system is measured, the only possible outcomes are the eigenvalues of the corresponding Hermitian operator (observable). Upon measurement, the system collapses into the eigenstate associated with the observed eigenvalue. The probability of obtaining a particular eigenvalue is given by the Born rule: it is the squared modulus of the inner product between the system's state and the corresponding eigenstate.*

Postulate 2.6 (Born Rule) *Let $\hat{O}$ be an observable with eigenstates $|\psi_i\rangle$ and eigenvalues o_i. If the system is in a normalized state $|\psi\rangle$, then the probability of obtaining the measurement outcome o_i is*

$$P(o_i) = |\langle\psi_i|\psi\rangle|^2.$$

The expectation value of $\hat{O}$ is therefore

$$\langle\hat{O}\rangle = \sum_i o_i P(o_i) = \sum_i o_i |\langle\psi_i|\psi\rangle|^2.$$

Using

$$|\langle\psi_i|\psi\rangle|^2 = \langle\psi|\psi_i\rangle\langle\psi_i|\psi\rangle,$$

we obtain

$$\langle\hat{O}\rangle = \sum_i o_i \langle\psi|\psi_i\rangle\langle\psi_i|\psi\rangle = \langle\psi|\left(\sum_i o_i |\psi_i\rangle\langle\psi_i|\right)|\psi\rangle.$$

Since

$$\hat{O} = \sum_i o_i |\psi_i\rangle\langle\psi_i|,$$

we get

$$\langle\hat{O}\rangle = \langle\psi|\hat{O}|\psi\rangle.$$

3 Schrödinger's Picture

Postulate 3.1 (Schrödinger Equation) *The general form of Schrödinger equation is*

$$i\hbar\frac{\partial}{\partial t}|\Psi\rangle = \hat{H}|\Psi\rangle$$

Definition 3.2 (Propagator) *Schrödinger equation provides the way system evolves in time, but we want a solution for the eqquation, so we introduce propagator $U(t, t_0)$ such that*

$$|\Psi(t)\rangle = U(t, t_0)|\Psi(t_0)\rangle.$$

Simply take a look at Schrödinger equation, it is a first order linear differential equation:

$$\frac{\partial}{\partial t}|\Psi(t)\rangle = -\frac{i}{\hbar}\hat{H}|\Psi(t)\rangle$$

For ODE in the form of

$$\frac{\mathrm{d}}{\mathrm{d}t}y(t) = Ay(t)$$

We have its solution in the form:

$$y(t) = e^{A(t-t_0)}y(t_0)$$

So we now have:

$$|\Psi(t)\rangle = U(t, t_0)|\Psi(t_0)\rangle$$

where

$$U(t, t_0) = e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}$$

An important property is the the propagator is a unitary transformation of $\mathcal{H}$:

$$U(t, t_0) \in U(\mathcal{H}).$$

Example 3.3 (Free Quantum Particle) *For some one-dimensional, non-relativistic free particle with mass m, since the Hamiltonian is in the following form*

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$$

we have the Schrödinger equation

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(x,t)$$

Because the free-particle spectrum is continuous, the momentum eigenstates are normalised using the Dirac delta distribution:

$$\langle k|k'\rangle = \int_{-\infty}^{\infty} \psi_k^*(x)\,\psi_{k'}(x)\,dx = \delta(k-k').$$

Substituting $\psi_k(x) = Ne^{ikx}$ gives

$$\langle k|k'\rangle = |N|^2\int_{-\infty}^{\infty} e^{i(k'-k)x}\,dx = |N|^2(2\pi)\,\delta(k'-k).$$

Thus,

$$|N|^2 = \frac{1}{2\pi} \qquad \Rightarrow \qquad N = \frac{1}{\sqrt{2\pi}}.$$

Therefore, the eigenstates are

$$\psi_k(x) = \frac{1}{\sqrt{2\pi}}\,e^{ikx}, \qquad k \in \mathbb{R}$$

And the eigenvalues are

$$E_k = \frac{\hbar^2 k^2}{2m}$$

Moreover, if it is three-dimensional, we have

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = -\frac{\hbar^2}{2m}\nabla^2\psi(x,t)$$

And the corresponding eigenstates are

$$\psi_{\mathbf{k}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}}\,e^{i\mathbf{k}\cdot\mathbf{x}}, \qquad \mathbf{k} \in \mathbb{R}^n$$

Example 3.4 (Quantum Harmonic Oscillator) *The Hamiltonian for QHC is*

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + \frac{1}{2}m\omega^2 x^2$$

So we have the Schrd̈inger equation in the following form

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + \frac{1}{2}m\omega^2 x^2\right]\psi(x,t)$$

The eigenvalues are

$$E_n = \hbar\omega(n + \frac{1}{2}), \qquad n = 0, 1, 2, 3, ...$$

The eigenstates are

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) e^{-\frac{m\omega x^2}{2\hbar}}$$

When $n = 0$

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}}$$

Usually, we use ladder operator: raising operator $\hat{a}^\dagger$ *and decreasing operator* $\hat{a}$ *defined as*

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right),$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)$$

They satisfy

$$[\hat{a}, \hat{a}^\dagger] = 1$$

The Hamiltonian is rewritten as

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$$

And there is this ladder-like relation

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle\,,$$

$$\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

Therefore, all states can be generated with the base state

$$|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n\,|0\rangle$$

Example 3.5 (Particle in a Box) *Assume we have a particle on x-axis, with zero potential between* $x = 0$ *and* $x = L$ *and infinite potential out of the box. The Hamiltonian of this particle is*

$$\hat{H} = \begin{cases} -\frac{\hbar^2}{2m}\nabla^2 + 0, & 0 < x < L, \\ +\infty, & \textit{otherwise.} \end{cases}$$

The corresponding eigenstates are

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L},\ n = 1, 2, 3, ...$$

The eigenvalues are

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

Example 3.6 (Hydrogen Atom) *The Hamiltonian is*

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

We solve the equation

$$\left(-\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}\right)\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

The main idea is separation of variables. Since the potential is only dependent on the magnitude and not on the direction of $\mathbf{r}$*, we use spherical coordinate and let*

$$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$$

We have

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin(\theta)}\frac{\partial}{\partial\theta}\left(\sin(\theta)\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2(\theta)}\frac{\partial^2}{\partial\phi^2}$$

and

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin(\theta)}\frac{\partial}{\partial\theta}\left(\sin(\theta)\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2(\theta)}\frac{\partial^2}{\partial\phi^2}\right]$$

We then rewrite the Hamiltonian in spherical coordinate

$$\hat{H} = \frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2\mu r^2} - \frac{e^2}{4\pi\epsilon_0 r}$$

The first term only consists of derivative of r *and second only consists of angular momentum* $\hat{L}$*.*

The second part satisfies

$$\hat{L}^2 Y_\ell^m(\theta,\phi) = \hbar^2\ell(\ell+1)Y_\ell^m(\theta,\phi)$$

and

$$\hat{L}_z Y_\ell^m(\theta,\phi) = \hbar m Y_\ell^m(\theta,\phi)$$

where

$$l = 0,1,2,\ldots \qquad m = -l,\ldots,l$$

The solution $Y_\ell^m(\theta,\phi)$ *is standing wave on a sphere.*

Now that we know the angular dependence, we can rewrite the wave function

$$\psi(r,\theta,\phi) = R(r)Y_\ell^m(\theta,\phi)$$

Sub back to the Schrödinger equation, the angular part cancels

$$-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{\hbar^2\ell(\ell+1)}{2\mu r^2}R - \frac{e^2}{4\pi\epsilon_0 r}R = ER$$

We let

$$R(r) = \frac{u(r)}{r}$$

And we get the effective potential

$$V_{\text{eff}}(r) = \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{e^2}{4\pi\epsilon_0 r}$$

The wave function is in the following form

$$\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r) Y_\ell^m(\theta,\phi)$$

and

$$n = 1, 2, 3, \ldots$$

$$\ell = 0, 1, \ldots, n-1$$

$$m = -\ell, -\ell+1, \ldots, \ell.$$

The eigenvalues are

$$E_n = -\frac{13.6eV}{n^2}, \qquad n = 1, 2, 3, ...$$

Now we will see what happens to quantum spin syetem.

For a single spin system, the Hilbert space is $\mathcal{H} = \mathbb{C}^2$. The state is a 2-dimensional complex vector:

$$|\psi(t)\rangle = \begin{pmatrix} \psi_1(t) \\ \psi_2(t) \end{pmatrix}$$

where $\||\psi(t)\rangle\|^2 = 1$

The general Hamiltonian for a quantum spin is:

$$H = a_0 I + \mathbf{a} \cdot \sigma = a_0 I + a_x \sigma_x + a_y \sigma_y + a_z \sigma_z$$

where I is the identity matrix, σ_x,σ_y and σ_z are Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

We now check a simplest case.

Example 3.7 (Constant z-field) *The Hamiltonian is*

$$\hat{H} = a_0 I + a\sigma_z$$

The eigenstates are

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

And the corresponding eigenvalues are:

$$E_+ = a_0 + a, \quad E_- = a_0 - a$$

The propagator is

$$U(t) = e^{-i\hat{H}t/\hbar} = \begin{pmatrix} e^{-i(a_0+a)t/\hbar} & 0 \\ 0 & e^{-i(a_0-a)t/\hbar} \end{pmatrix}$$

If the case is more general, for example, the Hamiltonian is in the following form

$$\hat{H} = a_0 I + \mathbf{a} \cdot \sigma$$

where

$$\mathbf{a} = a\mathbf{n}$$

and $\mathbf{n}$ is the unit vector describing the direction of $\mathbf{a}$.

Assume the unit vector is in the following form

$$\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$$

Then

$$\mathbf{n} \cdot \boldsymbol{\sigma} = \begin{pmatrix} \cos\theta & e^{-i\phi}\sin\theta \\ e^{i\phi}\sin\theta & -\cos\theta \end{pmatrix}$$

The eigenstates are

$$|\mathbf{n}, +\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} \end{pmatrix},$$

$$|\mathbf{n}, -\rangle = \begin{pmatrix} -\sin\frac{\theta}{2} \\ e^{i\phi}\cos\frac{\theta}{2} \end{pmatrix}$$

They satisfy

$$(\mathbf{n} \cdot \boldsymbol{\sigma})\,|\mathbf{n}, +\rangle = +\,|\mathbf{n}, +\rangle\,,$$

$$(\mathbf{n} \cdot \boldsymbol{\sigma})\,|\mathbf{n}, -\rangle = -\,|\mathbf{n}, -\rangle$$

This means that the eigenvalues are

$$a_0 \pm a$$

When calculating the propagator for quantum spin, we first substitute in the Hamiltonian and get

$$U(t) = e^{-ia_0 t/\hbar} e^{-ia(\mathbf{n}\cdot\boldsymbol{\sigma})t/\hbar}$$

We make use of this formula

$$e^{-i\theta\mathbf{n}\cdot\boldsymbol{\sigma}} = \cos\theta\, I - i\sin\theta\, \mathbf{n} \cdot \boldsymbol{\sigma}$$

And then we have the propagator

$$U(t) = e^{-ia_0 t/\hbar}\left[\cos\left(\frac{at}{\hbar}\right) I - i\sin\left(\frac{at}{\hbar}\right) \mathbf{n} \cdot \boldsymbol{\sigma}\right]$$

Now that we have seen the quantum spin, we will discuss the spin-1/2 charged particle. The Hilbert space is $\mathcal{H} = L^2(\mathbb{R}^3, \mathbb{C}^2)$.

Since now we have the particle in the sapce, the wave function must describe the spin at any point in space. So we have the wave function in the following form

$$\Psi(\mathbf{r}, t) = \begin{pmatrix} \psi_+(\mathbf{r}, t) \\ \psi_-(\mathbf{r}, t) \end{pmatrix}$$

where the two argument describe the amplitude of finding the spin up or down at $\mathbf{r}$.

The Pauli Hamiltonian is

$$\hat{H} = \frac{1}{2m}\big(\mathbf{p} - q\mathbf{A}\big)^2 - \frac{q\hbar}{2m}\,\boldsymbol{\sigma} \cdot \mathbf{B}$$

The first term describes the orbital motion under minimal coupling, while the last term describes the coupling between the spin magnetic moment and the external magnetic field.

Now we see the case of uniform magnetic field in z direction.

Example 3.8 (Uniform Magnetic Field in z Direction) *Now we have magnetic field*

$$\mathbf{B} = B\hat{z}$$

Choose Landau gauge

$$\mathbf{A} = (0, Bx, 0)$$

so that $\mathbf{B} = \nabla \times \mathbf{A}$

The Hamiltonian is now

$$\hat{H} = \frac{1}{2m}\left[p_x^2 + (p_y - qBx)^2 + p_z^2\right] - \frac{q\hbar B}{2m}\sigma_z$$

We will see how we deal with this in the remark.

The energy is

$$E_{n,k_z,s} = \hbar\omega_c\left(n + \frac{1}{2}\right) + \frac{\hbar^2 k_z^2}{2m} - s\frac{q\hbar B}{2m}, \qquad s = \pm 1$$

Remark 3.9 (Pauli Hamiltonian) *Since the spin-1/2 charged particle under a magnetic field in z direction is translationally invariant along y and z, take*

$$\Psi(\mathbf{r}) = e^{ik_y y}e^{ik_z z}\begin{pmatrix}\phi_\uparrow(x)\\ \phi_\downarrow(x)\end{pmatrix}.$$

Define

$$\omega_c = \frac{|q|B}{m}, \qquad \ell_B = \sqrt{\frac{\hbar}{|q|B}}, \qquad x_0 \equiv \frac{\hbar k_y}{qB}.$$

Then the orbital Hamiltonian becomes

$$H_{\text{orb}} = \frac{p_x^2}{2m} + \frac{1}{2}m\omega_c^2(x - x_0)^2 + \frac{\hbar^2 k_z^2}{2m}.$$

Thus the orbital energies are the Landau levels:

$$E_{n,k_z}^{(\text{orb})} = \hbar\omega_c\Big(n + \tfrac{1}{2}\Big) + \frac{\hbar^2 k_z^2}{2m}, \qquad n = 0, 1, 2, \ldots$$

with eigenfunctions

$$\phi_n(x - x_0) = \frac{1}{\sqrt{2^n n!}\,\sqrt{\pi}\,\ell_B}\exp\Big[-\frac{(x - x_0)^2}{2\ell_B^2}\Big]\,H_n\Big(\tfrac{x - x_0}{\ell_B}\Big).$$

When it comes to the Zeeman term

$$H_{\text{spin}} = -\frac{q\hbar B}{2m}\,\sigma_z, \qquad \sigma_z\chi_s = s\,\chi_s, \qquad s = \pm 1.$$

Hence

$$E_s^{(\text{spin})} = -\,s\,\frac{q\hbar B}{2m}.$$

So the total spectrum is

$$E_{n,k_z,s} = \hbar\omega_c\Big(n + \tfrac{1}{2}\Big) + \frac{\hbar^2 k_z^2}{2m} - s\,\frac{q\hbar B}{2m} \qquad (n = 0, 1, 2, \ldots,\ s = \pm 1)$$

The eigenstates are

$$\Psi_{n,k_y,k_z,s}(\mathbf{r}) = \frac{e^{ik_y y}e^{ik_z z}}{\sqrt{L_y L_z}}\,\phi_n(x - x_0)\,\chi_s, \qquad x_0 = \frac{\hbar k_y}{qB}.$$

The orbital part ϕ_n produces the Landau–level structure, and χ_s gives the Zeeman spin splitting.

4 Heisenberg Picture

While in the Schrödinger picture the state evolves in time and the observables are (usually) time-independent operators, in the Heisenberg picture, the state is fixed and the observable evolve in time.

Both pictures are physically equivalent.

In particular, expectation values are identical in the Schrödinger and Heisenberg pictures for any observable $A : \mathcal{H} \to \mathcal{H}$. Indeed by using (3.2), the property (3) and the definition

$$A^H(t, t_0) = U(t, t_0)^* A\, U(t, t_0)$$

we get

$$\langle A\rangle(t) := \underbrace{\langle\psi(t)|A|\psi(t)\rangle}_{\text{expect. val. in Schrödinger}} = \langle\psi(t_0)|U(t,t_0)^* A\, U(t,t_0)\,|\psi(t_0)\rangle = \underbrace{\langle\psi(t_0)|A^H(t)|\psi(t_0)\rangle}_{\text{expect. val. in Heisenberg}}.$$

Now we see the general case.

Theorem 4.1 (Heisenberg's Equation)

$$\frac{\mathrm{d}A_H}{\mathrm{d}t}(t) = \frac{i}{\hbar}[\hat{H}_H,\, A_H(t)]$$

We start from Schrödinger's equation:

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\,|\psi_S(t)\rangle = \hat{H}_S\,|\psi_S(t)\rangle$$

Plug in the propagator by using (3.2). Since this must be valid for all $\psi(t_0)$, we have:

$$\frac{\mathrm{d}U}{\mathrm{d}t} = -\frac{i}{\hbar}\hat{H}_S U$$

Now that we have $\frac{\mathrm{d}U}{\mathrm{d}t}$, naturally, we want to also have $\frac{\mathrm{d}U^\dagger}{\mathrm{d}t}$. We know that $U^\dagger U = \mathbb{I}$, so we differentiate both sides:

$$\frac{\mathrm{d}U^\dagger}{dt}U + U^\dagger\frac{\mathrm{d}U}{\mathrm{d}t} = 0.$$

After that, plug in the result for $\frac{\mathrm{d}U}{\mathrm{d}t}$ and rearrange the equation:

$$\frac{\mathrm{d}U^\dagger}{\mathrm{d}t} = -U^\dagger\frac{\mathrm{d}U}{\mathrm{d}t}U^\dagger = \frac{i}{\hbar}U^\dagger\hat{H}_S$$

The main idea of these work is to transfer the time dependence of expectation from the state of an object to the observable, so that even if we only know the state of an object at some time, we can observe it at some other time. For the expectation value:

$$\begin{aligned}\langle A\rangle(t) &= \langle\psi_S(t)|A_S|\psi_S(t)\rangle \\ &= \left\langle U^\dagger(t,t_0)\right\rangle A_S U(t,t_0)\psi_H \\ &= \langle\psi_H|A_H(t)|\psi_H\rangle\end{aligned}$$

where we used (4):

$$A_H(t) = U^\dagger(t, t_0) A_S U(t, t_0)$$

Differentiate both sides:

$$\frac{\mathrm{d}A_H(t)}{\mathrm{d}t} = \frac{\mathrm{d}U^\dagger(t, t_0)}{\mathrm{d}t} A_S U(t, t_0) + U^\dagger(t, t_0) \frac{\mathrm{d}A_S}{\mathrm{d}t} U(t, t_0) + U^\dagger(t, t_0) r A_S \frac{\mathrm{d}U(t, t_0)}{\mathrm{d}t}$$

Substitute $\frac{\mathrm{d}U^\dagger}{\mathrm{d}t}$ and $\frac{\mathrm{d}U}{\mathrm{d}t}$ in:

$$\frac{\mathrm{d}A_H(t)}{\mathrm{d}t} = \frac{i}{\hbar} U^\dagger(t, t_0) \hat{H}_S A_S U(t, t_0) + U^\dagger(t, t_0) \frac{\mathrm{d}A_S}{\mathrm{d}t} U(t, t_0) - \frac{i}{\hbar} U^\dagger(t, t_0) A_S \hat{H}_S U(t, t_0)$$

Rearrange the equation, we have:

$$\frac{\mathrm{d}A_H(t)}{\mathrm{d}t} = \frac{i}{\hbar} U^\dagger(t, t_0) [\hat{H}_S,\ A_S] U(t, t_0) + U^\dagger(t, t_0) \frac{\mathrm{d}A_S}{\mathrm{d}t} U(t, t_0)$$

Now, since A_S is not dependent on time, $\frac{\mathrm{d}A_S}{\mathrm{d}t} = 0$,

$$\begin{aligned}
\frac{\mathrm{d}A_H(t)}{\mathrm{d}t} &= \frac{i}{\hbar} U^\dagger(t, t_0) [\hat{H}_S,\ A_S] U(t, t_0) \\
&= \frac{i}{\hbar} (U^\dagger(t, t_0) \hat{H}_S A_S U(t, t_0) - U^\dagger(t, t_0) A_S \hat{H}_S U(t, t_0)) \\
&= \frac{i}{\hbar} (U^\dagger(t, t_0) \hat{H}_S U(t, t_0) U^\dagger(t, t_0) A_S U(t, t_0) - U^\dagger(t, t_0) A_S U(t, t_0) U^\dagger(t, t_0) \hat{H}_S U(t, t_0)) \\
&= \frac{i}{\hbar} (\hat{H}_H(t) A_H(t) - A_H(t) \hat{H}_H(t)) \\
&= \frac{i}{\hbar} [\hat{H}_H(t),\ A_H(t)]
\end{aligned}$$

When solving problems with Heisenberg's equation, we follow the 4 steps: 1. Decide what observable A we want to use; 2. Write the Hamiltonian H for the system; 3. Calculate the commutator $[H,\ A]$; 4. Rearrange the equation.

Let us see some examples.

Example 4.2 (Quantum Particle in a Potential) *First, let's find the velocity of this particle. Set*

$$A = \hat{x} = x$$

The Hamiltonian operator for this system is:

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x)\mathbb{I}$$

Calculate the commutator:
Note that the operators $V\mathbb{I}$ and $\hat{x}$ commute: $[V\mathbb{I}, \hat{x}] = 0$, so:

$$\frac{\mathrm{d}\hat{x}(t)}{\mathrm{d}t} = \frac{i}{\hbar} \left[\frac{\hat{p}(t)^2}{2m} + V\mathbb{I},\ \hat{x} \right] = \frac{i}{\hbar} \left[\frac{\hat{p}^2}{2m}\, \hat{x} \right]$$

Remember $\left[A^2,\, B\right] = A\,[A,\, B] + [A,\, B]\,A$*:*

$$\frac{\mathrm{d}\hat{x}}{\mathrm{d}t} = \frac{i}{2\hbar m}\left(\hat{p}\,[\hat{p},\, \hat{x}] + [\hat{p},\, \hat{x}]\,\hat{p}\right) = \frac{\hat{p}}{m}$$

where we used $[\hat{x}(t), \hat{p}(t)] = i\hbar\mathbb{I}$*.*

This is the same as the result in Newtonian mechanics.

Now, let's find the momentum and check if it gives the same result as in Newtonian mechanics. Set

$$A = \hat{p} = -i\hbar\nabla$$

The Hamiltonian operator for the system:

$$H = \frac{\hat{p}^2}{2m} + V(x)\mathbb{I}$$

Then calculate the commutator:

$$\frac{\mathrm{d}\hat{p}}{\mathrm{d}t} = \frac{i}{\hbar}\left[\frac{\hat{p}^2}{2m} + V\mathbb{I},\, \hat{p}\right] = \frac{i}{\hbar}\left[\frac{\hat{p}^2}{2m},\, \hat{p}\right] + \frac{i}{\hbar}\left[V\mathbb{I},\, \hat{p}\right]$$

Note that $\left[\frac{\hat{p^2}}{2m},\, \hat{p}\right] = 0$ *and* $[V\mathbb{I},\, \hat{p}] = i\hbar\dfrac{\mathrm{d}V}{\mathrm{d}x}\mathbb{I}$*(Explanation in remark):*

$$\frac{\mathrm{d}\hat{p}}{\mathrm{d}t} = \frac{i}{\hbar}\cdot i\hbar\frac{\mathrm{d}V}{\mathrm{d}x} = -\frac{\mathrm{d}V}{\mathrm{d}x}$$

This is Newton's second law.

Remark 4.3 (Commutator Relation)

$$\left[\frac{\hat{p}^{\,2}}{2m},\, \hat{p}\right] = \frac{1}{2m}[\hat{p}^{\,2}, \hat{p}] = \frac{1}{2m}(\hat{p}^{\,2}\hat{p} - \hat{p}\hat{p}^{\,2}) = 0,$$

since $\hat{p}^{\,2}\hat{p} = \hat{p}\hat{p}^{\,2}$*, i.e. momentum operators commute with themselves.*

$$[V(\hat{x}), \hat{p}]\psi = V(-i\hbar\partial_x\psi) + i\hbar\partial_x(V\psi) = i\hbar(\partial_x V)\psi,$$

therefore

$$[V(\hat{x}), \hat{p}] = i\hbar\,\frac{dV}{dx}\mathbb{I}$$

Another useful relation:

$$\begin{aligned}
[\hat{p}, \hat{x}]\psi = \hat{p}(\hat{x}\psi) - \hat{x}(\hat{p}\psi) &= -i\hbar\,\partial_x(x\psi) + i\hbar x\,\partial_x\psi, \\
&= -i\hbar(\psi + x\partial_x\psi) + i\hbar x\partial_x\psi = -i\hbar\,\psi, \\
\implies [\hat{p}, \hat{x}] &= -i\hbar \quad \textit{(equivalently } [\hat{x}, \hat{p}] = i\hbar\textit{)}
\end{aligned}$$

Example 4.4 (Quantum Spin) *Let*

$$A = \hat{S}_i : \mathbb{C}^2 \to \mathbb{C}^2$$

be the observable given by:

$$(\hat{S}_i)_H(t) = e^{\frac{i}{\hbar}Ht}$$

Hamiltonian operator (Zeeman effect):

$$\hat{H} = -\gamma \hat{\mathbf{S}} \cdot \mathbf{B} = -\gamma \sum_j \hat{S}_j B_j : \mathbb{C}^2 \to \mathbb{C}^2$$

where

$$\hat{\mathbf{S}} = \frac{h}{2}(\sigma_1, \sigma_2, \sigma_3)$$

is the spin operator and $\mathbf{B}$ *is the external magnetic field vector.*

Note that $\left[(\hat{S}_i)_H, (\hat{S}_j)_H\right] = i\hbar\varepsilon_{ijk}\hat{S}_k$*:*

$$\begin{aligned}
\frac{\mathrm{d}\hat{S}_i}{\mathrm{d}t} &= \frac{i}{\hbar}\left[\hat{H}, (\hat{S}_i)_H\right] \\
&= \frac{i}{\hbar}\left[-\gamma \sum_j (\hat{S}_j)_H B_j, (\hat{S}_i)_H\right] \\
&= -\frac{i}{\hbar}\gamma \sum_j B_j \left[(\hat{S}_j)_H, (\hat{S}_i)_H\right] \\
&= \gamma \sum_j \varepsilon_{jik} B_j \hat{S}_k \\
&= \gamma \left(\hat{\mathbf{S}} \times \mathbf{B}\right)_i
\end{aligned}$$

So, the Heisenberg equation for the quantum spin is:

$$\frac{\mathrm{d}\hat{\mathbf{S}}}{\mathrm{d}t} = \gamma \hat{\mathbf{S}} \times \mathbf{B}$$

Example 4.5 (Spin-$1/2$ Charged Particle in an Electromagnetic Field) *For a spin-*$1/2$ *charged particle in an electromagnetic field, the Hamiltonian operator includes kinetic energy, potential energy and Zeeman effect:*

$$\hat{H} = \frac{1}{2m}\left(\hat{\mathbf{p}} - q\mathbf{A}(\hat{x})\right)^2 + q\phi(\hat{x}) - \gamma \hat{\mathbf{S}} \cdot \mathbf{B}(\hat{x}) : L^2\mathbb{R}^n, \mathbb{C}^2) \to L^2\mathbb{R}^n, \mathbb{C}^2).$$

Now, here's one thing we need to know: the first and second term describe the particle's motion in the space, while the third one describes the internal degree of freedom, i.e. the spin. It is basically combining the two example we mentioned above.

Here we introduce

$$\hat{\mathbf{\Pi}} = \hat{\mathbf{p}} - q\mathbf{A}\left(\hat{x}\right)$$

We have

$$\frac{\mathrm{d}\hat{x}}{\mathrm{d}t} = \frac{1}{m}\left(\hat{\mathbf{p}} - q\mathbf{A}\left(\hat{x}\right)\right)$$

and

$$\begin{aligned}
\frac{d\hat{\Pi}_i}{dt} &= \frac{q}{2m}\left[(\hat{\mathbf{\Pi}} \times \mathbf{B})_i - (\mathbf{B} \times \hat{\mathbf{\Pi}})_i\right] + qE_i(\hat{\mathbf{x}}) \\
\frac{\mathrm{d}\hat{\mathbf{p}}_i}{\mathrm{d}t} &= \frac{q}{2m}\left\{\hat{\Pi}_j, \nabla_i \mathbf{A}_j(\hat{x})\right\} + q\mathbf{E}_i(\hat{x}) + \gamma\nabla_i\left(\hat{\mathbf{S}} \cdot \mathbf{B}(\hat{x})\right)
\end{aligned}$$

with the classical counterpart

$$\dot{\Pi}_i = q(\dot{x} \times \mathbf{B})_i + qE_i = \frac{q}{m}(\Pi \times \mathbf{B})_i + qE_i$$

$$\dot{p}_i = q\dot{x}_j \nabla_i A_j(x) + qE_i = \frac{q}{m}\Pi_j \nabla_i A_j(x) + qE_i$$

When we calculate the spin precession, we again use the following relation

$$\left[\hat{S}_i, \hat{S}_j\right] = i\hbar\,\epsilon_{ijk}\hat{S}_k$$

and get

$$\frac{d\hat{\mathbf{S}}}{dt} = \gamma\hat{\mathbf{S}} \times \mathbf{B}$$

5 Path Integral

Definition 5.1 (Action) *For an object moving on a path from a to b, the action functional is defined as*

$$S[q] = \int_a^b L(t, q, \dot{q})dx$$

In classical mechanics, the object always takes the path with the minimal change in action.

Theorem 5.2 (Euler-Lagrange Equation) *For the given action functional, a path is the stationary (with respect to a small perbutation in q) if and only if*

$$\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{dt}}\left(\frac{\partial L}{\partial \dot{q}}\right) = 0$$

The Hamiltonian way of see this is different. We don't want to see $\dot{q}$. So we define the following.

Definition 5.3 (Hamiltonian)

$$H(\mathbf{q}, \mathbf{p}, t) = \sum_i p_i \dot{q}_i - L(\mathbf{q}, \dot{\mathbf{q}}, t)$$

where p_i is defined as: $\frac{\partial L}{\partial q_i}$.

And we have the following relation describing the motion of objects.

Theorem 5.4 (Hamiltonian equation)

$$\dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

With this, we can express the state of an object with a phase diagram, with general coordinate and general momentum being the two axis.

Now, we go to the formal definition and computation of the propagator via path integral.

Definition 5.5 (Amplitude (phase-space formulation)) *An object starts at (t_i, x_i), the amplitude of finding it at (t_f, x_f) is*

$$K_{fi} = A \sum_{paths\ x(t)} e^{iSx/\hbar}$$

The probability of finding the object at (t_f, x_f) is

$$\text{Prob}(x_f) = |K_{fi}|^2 \mathrm{d}x_f$$

Since the set of all paths is not a discrete set, we rewrite it as an integral.

$$K_{fi} = \int_{x_i}^{x_f} \mathcal{D}x \Phi[x] = \int_{x_i}^{x_f} \mathcal{D}x e^{iSx/\hbar}$$

It means we do a sum of all paths associated with a phase.
We use the trick of cutting the time into infinite slices to do this integral.

$$\int_{x_I}^{x_f} \mathcal{D}x \Phi[x] = \lim_{N\to\infty} A \int_{-\infty}^{\infty} \mathrm{d}x_1 \mathrm{d}x_2 ... \mathrm{d}x_{N-1} \Phi[x]$$

The above definition is the **phase-space formulation**.

Definition 5.6 (Amplitude (configuration formulation))

$$K(x, t; x_0, t_0) = \langle x|U(t, t_0)|x_0\rangle$$

The properties are closely related to those of $U(t, t_0)$.
For example,

$$K(x, t_0; x_0, t_0) = \langle x|x_0\rangle = \delta(x - x_0).$$

Example 5.7 (The Free Quantum Particle) *In this case, the action functional is*

$$S[x] = \int_{t_i}^{t_f} \mathrm{d}t \left\{ \frac{1}{2} m \left(\frac{\mathrm{d}x}{\mathrm{d}t} \right)^2 - U(x(t)) \right\}$$

Note that for a free particle, $U = 0$.
We take

$$\Delta t = \frac{t_f - t_i}{N}$$

Then we can work out the total action in the following form.

$$S = \frac{m}{2\Delta t} \left\{ (x_1 - x_i)^2 + (x_2 - x_1)^2 + ... + (x_f - x_{N-1})^2 \right\}$$

Plug into the path integral.

$$\int_{x_i}^{x_f} \mathcal{D}x\, e^{iS/\hbar} = \lim_{N\to\infty} A \int_{-\infty}^{\infty} dx_1\, dx_2 \cdots dx_{N-1}\, e^{\frac{i}{\hbar}\frac{m}{2\Delta t}\left[(x_1-x_i)^2+(x_2-x_1)^2+\cdots+(x_f-x_{N-1})^2\right]}$$

By working out the first two terms,

$$\int_{-\infty}^{\infty} dx_1\, e^{\frac{i}{\hbar}\frac{m}{2\Delta t}\left[(x_1-x_i)^2+(x_2-x_1)^2\right]} = \sqrt{\frac{\pi i\hbar\Delta t}{m}}\, e^{\frac{i}{\hbar}\frac{m}{4\Delta t}(x_2-x_i)^2}$$

We see a pattern, and therefore work out the entire path integral.

$$\int_{x_i}^{x_f} \mathcal{D}x\, e^{\frac{i}{\hbar}S} = \lim_{N\to\infty} A\left(\sqrt{\frac{2\pi i\hbar\Delta t}{m}}\right)^N \sqrt{\frac{m}{2\pi i\hbar\,(t_f-t_i)}}\, e^{\frac{i}{\hbar}\frac{1}{2}m\frac{(x_f-x_i)^2}{t_f-t_i}}$$

Now we see that all dependence on N is on $\left(\sqrt{\frac{2\pi i\hbar\Delta t}{m}}\right)^N$. To normalize the term, we take

$$A = \left(\sqrt{\frac{m}{2\pi i\hbar\Delta t}}\right)^N$$

Therefore, we have the result of the path integral, the amplitude

$$K_{fi} = \sqrt{\frac{m}{2\pi i\hbar\,(t_f-t_i)}}\, e^{\frac{i}{\hbar}\frac{1}{2}m\frac{(x_f-x_i)^2}{t_f-t_i}}$$

This is the quantum amplitude for a free particle in one dimension to propagate from x_i to x_f in the time interval from t_i to t_f.

We can easily generalise it to three dimensional space.

$$K_{fi} = \left(\frac{m}{2\pi i\hbar\,(t_f-t_i)}\right)^{3/2}\, e^{\frac{i}{\hbar}\frac{1}{2}m\frac{(x_f-x_i)^2}{t_f-t_i}}$$

Example 5.8 (The Quantum Particle in a Potential) *We already know that*

$$K_{fi} = \int_{x_i}^{x_f} \mathcal{D}x e^{iSx/\hbar}$$

When we have Lagrangian in the form

$$L(x,\dot{x}) = \frac{1}{2}m\dot{x}^2 - V(x)$$

We have the formal definition of the propagator with potential

$$K_{fi} = \int_{x_i}^{x_f} \mathcal{D}x \exp\left[\frac{i}{\hbar}\int_{t_i}^{t_f}\left(\frac{1}{2}m\dot{x}^2 - V(x)\right)\mathrm{d}t\right]$$

Usually, it is very hard to compute, because the integral isn't necessarily computable due to the form of the potential. For example, if the potential is in the form of $V = kx^4$, we cannot reduce it to the Gaussian integral. But we can still do some simple cases.

Let's see a simple example of a constant force field.

Example 5.9 (V(x)=-fx) *Consider a particle in a linear potential*

$$V(x) = -fx$$

where f is a constant force. The Lagrangian is

$$L(x,\dot{x}) = \frac{m}{2}\dot{x}^2 + fx$$

The propagator is formally given by

$$K(x_f, t_f; x_i, t_i) = \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x(t)\, \exp\left[\frac{i}{\hbar}\int_{t_i}^{t_f}\left(\frac{m}{2}\dot{x}^2 + fx\right)dt\right]$$

Let

$$T = t_f - t_i$$

For simplicity, set $t_i = 0$ and $t_f = T$. The classical equation of motion is

$$m\ddot{x} = f$$

The classical path satisfying

$$x_{\mathrm{cl}}(0) = x_i, \qquad x_{\mathrm{cl}}(T) = x_f$$

is

$$x_{\mathrm{cl}}(t) = x_i + \left(\frac{x_f - x_i}{T} - \frac{fT}{2m}\right)t + \frac{f}{2m}t^2$$

Substituting this into the action gives

$$S_{\mathrm{cl}} = \frac{m(x_f - x_i)^2}{2T} + \frac{fT(x_f + x_i)}{2} - \frac{f^2T^3}{24m}$$

Since this Lagrangian is quadratic, the path integral can be evaluated exactly. The result is

$$K(x_f, T; x_i, 0) = \left(\frac{m}{2\pi i\hbar T}\right)^{1/2} \exp\left\{\frac{i}{\hbar}\left[\frac{m(x_f - x_i)^2}{2T} + \frac{fT(x_f + x_i)}{2} - \frac{f^2T^3}{24m}\right]\right\}$$

Restoring $T = t_f - t_i$, this gives the propagator for the linear potential $V(x) = -fx$. When $f \to 0$, it reduces to the free-particle propagator.

When it comes to the quantum spin system, we must take into account of berry phase.

Definition 5.10 (Berry Phase) *The Berry phase is a geometric phase picked up when a quantum system is adiabatically carried around a closed path. In contrast to the dynamical phase, it depends on the path's geometry rather than on energy and time. For spin-1/2, spin directions form the Bloch sphere S^2, so the spin path integral sums over paths on this sphere and includes a Berry phase term.*

Definition 5.11 (Spin Coherent State) *A spin coherent state is the quantum state that most closely corresponds to a classical spin oriented in a well-defined direction. For a spin-s system, the spin coherent state directed along the unit vector*

$$\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$$

can be written as

$$|\mathbf{n}\rangle = e^{-i\phi\hat{S}_z/\hbar}e^{-i\theta\hat{S}_y/\hbar}\,|s, s\rangle$$

The spin coherent state $|\mathbf{n}\rangle$ is constructed by starting from the highest-weight state $|s, s\rangle$, whose spin is fully aligned along the $+z$ direction, and rotating it to an arbitrary direction $\mathbf{n}$.

When $s = 1/2$, it is given by

$$|\mathbf{n}\rangle = \cos\frac{\theta}{2}\,|\uparrow\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|\downarrow\rangle$$

It satisfies

$$\langle \mathbf{n} | \, \boldsymbol{\sigma} \, | \mathbf{n} \rangle = \mathbf{n}$$

so the expectation value of the spin acts as a classical vector oriented along $\mathbf{n}$*.In the spin path integral, one therefore sums over all possible paths* $\mathbf{n}(t)$ *on the Bloch sphere rather than over position paths* $x(t)$*.*

Theorem 5.12 (Completeness Relation)

$$\frac{2s+1}{4\pi} \int d\Omega \; |\mathbf{n}\rangle \langle \mathbf{n}| = 1$$

where

$$d\Omega = \sin\theta \, d\theta \, d\phi$$

This identity shows that the spin coherent states $|\mathbf{n}\rangle$ *are overcomplete: their projectors, integrated over all directions, reproduce the identity operator.*

The prefactor $\frac{2s+1}{4\pi}$ *is the normalization factor, where* $2s+1$ *is the dimension of the spin-s Hilbert space.*

Example 5.13 (Quantum Spin and Spin-$1/2$**)** *We want to calculate*

$$K(\mathbf{n}_f, t_f; \mathbf{n}_i, t_i) = \langle \mathbf{n}_f | e^{-\frac{i}{\hbar}\hat{H}(t_f - t_i)} | \mathbf{n}_i \rangle$$

To construct the path integral, we divide the time interval $T = t_f - t_i$ *into* N *small pieces, with*

$$\Delta t = \frac{T}{N}$$

Then

$$K(\mathbf{n}_f, t_f; \mathbf{n}_i, t_i) = \langle \mathbf{n}_f | \left(e^{-\frac{i}{\hbar}\hat{H}\Delta t} \right)^N | \mathbf{n}_i \rangle$$

At each intermediate time slice, we insert the completeness relation

$$\mathbf{1} = \frac{2s+1}{4\pi} \int d\Omega_k \; |\mathbf{n}_k\rangle \langle \mathbf{n}_k|$$

Thus the propagator becomes

$$K = \lim_{N\to\infty} \prod_{k=1}^{N-1} \left[\frac{2s+1}{4\pi} \int d\Omega_k \right] \prod_{k=0}^{N-1} \langle \mathbf{n}_{k+1} | e^{-\frac{i}{\hbar}\hat{H}\Delta t} | \mathbf{n}_k \rangle$$

where

$$\mathbf{n}_0 = \mathbf{n}_i, \qquad \mathbf{n}_N = \mathbf{n}_f$$

For small Δt*, we approximate*

$$\langle \mathbf{n}_{k+1} | e^{-\frac{i}{\hbar}\hat{H}\Delta t} | \mathbf{n}_k \rangle \approx \langle \mathbf{n}_{k+1} | \mathbf{n}_k \rangle \exp\left[-\frac{i}{\hbar} H(\mathbf{n}_k) \Delta t \right]$$

where

$$H(\mathbf{n}) = \langle \mathbf{n} | \hat{H} | \mathbf{n} \rangle$$

The first factor $\langle \mathbf{n}_{k+1} | \mathbf{n}_k \rangle$ *is not a dynamical contribution. It comes from the change of the spin direction on the Bloch sphere and gives the Berry phase.*

In the continuum limit,

$$\prod_{k=0}^{N-1} \langle \mathbf{n}_{k+1} | \mathbf{n}_k \rangle = \exp\left[\frac{i}{\hbar}\int_{t_i}^{t_f} dt\, \hbar s(1-\cos\theta)\dot{\phi}\right]$$

Therefore the spin path integral is

$$K(\mathbf{n}_f, t_f; \mathbf{n}_i, t_i) = \int_{\mathbf{n}(t_i)=\mathbf{n}_i}^{\mathbf{n}(t_f)=\mathbf{n}_f} \mathcal{D}\Omega\, e^{\frac{i}{\hbar}S[\theta,\phi]}$$

with action

$$S[\theta,\phi] = \int_{t_i}^{t_f} dt\, \Big[\hbar s(1-\cos\theta)\dot{\phi} - H(\theta,\phi)\Big]$$

Here

$$\mathcal{D}\Omega = \lim_{N\to\infty} \prod_{k=1}^{N-1} \frac{2s+1}{4\pi} \sin\theta_k\, d\theta_k\, d\phi_k$$

For a spin-1/2 particle,

$$s = \frac{1}{2}$$

so the action becomes

$$S[\theta,\phi] = \int_{t_i}^{t_f} dt\, \left[\frac{\hbar}{2}(1-\cos\theta)\dot{\phi} - H(\theta,\phi)\right]$$

where the first term is the Berry phase term and the second term is the usual dynamical contribution.

6 Case Study: Multiparticle

Definition 6.1 (Multiparticle Wave Function) *If a system has more than one particle, the wave function is in the following form*

$$\Psi(x_1, x_2, x_3, \ldots, x_N, t)$$

The meaning is the associated amplitude at time t and particle 1 at x_1, 2 at x_2 etc..
Therefore, the probability density is

$$|\Psi(x_1, x_2, x_3, \ldots, x_N, t)|^2$$

More specifically,

$$|\Psi(x_1, x_2, x_3, \ldots, x_N, t)|^2 dx_1 dx_2 \ldots dx_N$$

describes the probability at time t of particle 1 at $[x_1, x_1 + dx_1]$, 2 at $[x_2, x_2 + dx_2]$, ... , N at $[x_N, x_N + dx_N]$.

Definition 6.2 (Tensor Product) *For one particle whose Hilbert space is $\mathcal{H}$, if we have many of them, say N, then the Hilbert space of the system is*

$$\mathcal{H}^{\otimes N} = \underbrace{\mathcal{H}\otimes\mathcal{H}\otimes\cdots\otimes\mathcal{H}}_{N \text{ times}}$$

Postulate 6.3 (Multiparticle Schrödinger Equation) *A multiparticle system follows Schrödinger equation*

$$i\hbar\frac{\partial}{\partial t}\Psi(x_1, x_2, \dots, x_N, t) = \hat{H}\Psi(x_1, x_2, \dots, x_N, t)$$

An example for the Hamiltonian is

$$\hat{H} = \sum_{i=1}^{N}\left[-\frac{\hbar^2}{2m_i}\nabla_i^2 + V_{\text{ext}}(\mathbf{x}_i)\right] + \sum_{i<j}U(\mathbf{x}_i - \mathbf{x}_j)$$

a sum of the kinetic energy, the external potential energy and internal potential energy.

Definition 6.4 (Identical Particles) *For a system of two particles, if we have two distinguishable particles, for example an electron and a proton, the probability before and after exchange them should be different*

$$|\Psi(x_1, x_2)|^2 \neq |\Psi(x_2, x_1)|^2$$

But if we have two identical particles, the probability should be the same,

$$|\Psi(x_1, x_2)|^2 = |\Psi(x_2, x_1)|^2$$

which then gives us

$$\Psi(x_1, x_2) = \pm\Psi(x_2, x_1)$$

When the positive sign appears, it's called symmetric wave function and when the negative sign appears, it's called antisymmetric wave function.

Definition 6.5 (Bosons) *The wave function of bosons does not change before and after swapping two particles*

$$\Psi(\dots, x_i, \dots, x_f, \dots) = \Psi(\dots, x_f, \dots, x_i, \dots)$$

For example, photons are bosons.

Definition 6.6 (Fermions) *After swapping two particles, the sign of the wave function changes*

$$\Psi(\dots, x_i, \dots, x_f, \dots) = -\Psi(\dots, x_f, \dots, x_i, \dots)$$

If there are two fermions in the exact same state, swapping them does not change the wave function, then the property of fermions gives

$$\Psi = -\Psi$$

so

$$\Psi = 0$$

This is the Pauli exclusion principle: two fermions cannot stay in the same quantum state.

Some examples of fermions: electrons, protons, neutrons.

Theorem 6.7 (Multiparticle Heisenberg Equation) *The equation is the same as the usual Heisenberg equation only substituting $\hat{H}$ with $\hat{H}_N$*

$$\frac{\mathrm{d}A_H(t)}{\mathrm{d}t} = \frac{i}{\hbar}\left[\hat{H_N}, \hat{A}_H\right] + \left(\frac{\partial\hat{A}_S}{\partial t}\right)_H$$

So when A_S is not dependent on time, the Heisenberg equation is

$$\frac{\mathrm{d}A_H(t)}{\mathrm{d}t} = \frac{i}{\hbar}\left[\hat{H_N}, \hat{A}_H\right]$$

Theorem 6.8 (Path Integral Formulation for Multiparticle system) *The path integral formulation is generalised in a similar way. Instead of summing over a single path, we sum over all possible paths of N particles*

$$K_N = \int \mathcal{D}x_1(t) \cdots \mathcal{D}x_N(t) \exp\left[\frac{i}{\hbar} S_N[x_1, \ldots, x_N]\right]$$

More explicitly,

$$K_N = \int_{x_1(t_i)=x_1^i}^{x_1(t_f)=x_1^f} \mathcal{D}x_1(t) \cdots \int_{x_N(t_i)=x_N^i}^{x_N(t_f)=x_N^f} \mathcal{D}x_N(t) \, \exp\left[\frac{i}{\hbar} S_N[x_1, \ldots, x_N]\right]$$

For the system with internal potential $U(x_i - x_j)$, the action can be written as

$$S_N[x_1, \ldots, x_N] = \int_{t_i}^{t_f} \left\{ \sum_{i=1}^{N} \left[\frac{1}{2} m_i \dot{x}_i^2 - V(x_i)\right] - \sum_{i<j} U(x_i - x_j) \right\} dt$$

Now, we have the theory to discribe a system with N fixed number of particles, naturally we will generalise it to arbitrary number.

When there are arbitrary number of particles, we use the following to describe the system

$$|\Psi\rangle = c_0 |\Psi_0\rangle + c_1 |\Psi_1\rangle + c_2 |\Psi_2\rangle + \ldots$$

where

$$|\Psi_N\rangle \in \mathcal{H}_N$$

Here, $|\Psi_0\rangle$ is the state where the system has zero particle, $|\Psi_1\rangle$ is the state where the system has one particle, etc..

The coefficient c_N is the probability amplitude, and they satisfy

$$\sum_{N=0}^{\infty} |c_N|^2 = 1$$

Definition 6.9 (Creation and Annihilation Operator) *The annihilation operator is written as*

$$\hat{a}$$

and the creation operator is written as

$$\hat{a}^\dagger$$

They satisfy

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle$$

and

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

where $|n\rangle$ means the state with n particles.

The maening of creation and annihilation operator is changing the number of particles in the system.

Definition 6.10 (Particle Number Operator) *The particle number operator is defined as*

$$\hat{N} = \hat{a}^\dagger \hat{a}$$

When it acts on $|n\rangle$, the annihilation oeprator first acts on the state

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$$

and then we let the creation operator to act on the state

$$\hat{a}^\dagger \sqrt{n}\,|n-1\rangle = \sqrt{(n-1)+1}\sqrt{n}\,|n\rangle = n\,|n\rangle$$

so the eigenstate is $|n\rangle$ and the eigenvalue is the number of particles.

Remark 6.11 (What does it mean to creat a particle) *Generally, if there are many possible fixed number particle mode, we express it in the following form*

$$|n_1, n_2, n_3, \ldots\rangle$$

where n_i indicates the number of particles in the ith mode.

For example we have the state

$$|0, 0, 0, \ldots\rangle$$

which means that there is no particle in each mode.

If we let the creation operator act on the state, it becomes

$$\hat{a}_2^\dagger\,|0, 0, 0, \ldots\rangle = |0, 1, 0, \ldots\rangle$$

It means that there is one particle in the second mode.

If we have creation operator act on it again

$$\hat{a}_2^\dagger\,|0, 1, 0, \ldots\rangle = \sqrt{2}\,|0, 2, 0, \ldots\rangle$$

We now have two bosons in the second mode. They are bosons because two fermions are not allowed to occupy the same mode.

Definition 6.12 (Fork Space) *The Fork space is defined as*

$$\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}_N$$

Previously we have

$$|\Psi\rangle = \begin{pmatrix} |\Psi_0\rangle \\ |\Psi_1\rangle \\ |\Psi_2\rangle \\ \cdots \end{pmatrix}$$

where

$$|\Psi_N\rangle \in \mathcal{H}_N$$

The Fork space can be viewed as the Hilbert space that hosts $|\Psi\rangle$

$$\mathcal{F} = \mathcal{H}_\prime \oplus \mathcal{H}_\infty \oplus \mathcal{H}_\in \ldots$$

where each layer corresponds a circumstance with a certain number of particles.

Example 6.13 (Vacuum Sector) *Note that each specific Hilbert space is called a sector, for example, the vacuum sector is*

$$\mathcal{H}_0$$

it means that there is no such particles in the system

$$\hat{N}\left|0\right\rangle = 0$$

Remark 6.14 (Vacuum Sector) *This vacuum sector does not mean there is nothing and there can be some nontrivial properties according to quantum field theory, which we will not discuss.*

Example 6.15 (One-Particle Sector) *The one-particle sector is*

$$\mathcal{H}_1$$

and a state in this sector is

$$\left|\Psi_1\right\rangle = \int dx \psi_1(x) \left|x\right\rangle$$

Example 6.16 (A State with Arbitrary Number of Particles) *Consider the state*

$$\left|\Psi\right\rangle = \frac{1}{2}\left|0\right\rangle + \frac{1}{\sqrt{2}}\left|1\right\rangle + \frac{1}{2}\left|2\right\rangle$$

and assume

$$\langle\Psi_1|\Psi_1\rangle = \langle\Psi_2|\Psi_2\rangle = 1$$

We have

$$P(N=0) = \frac{1}{4}$$

$$P(N=1) = \frac{1}{2}$$

$$P(N=2) = \frac{1}{4}$$

Remark 6.17 (Particle Number Operator) *Recall the concept of particle number operator, it measures the number of particles in a system.*

Assume all $\left|\Psi_N\right\rangle$ are normalised, we have

$$\left\langle \hat{N} \right\rangle = \sum_{N=0}^{\infty} N|c_N|$$

For the previous example

$$\left|\Psi\right\rangle = \frac{1}{2}\left|0\right\rangle + \frac{1}{\sqrt{2}}\left|1\right\rangle + \frac{1}{2}\left|2\right\rangle$$

we have

$$\left\langle \hat{N} \right\rangle = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1$$

It means that the expectation value of number of particles is 1 *but there could be cases with* 0, 1 *or* 2 *particles.*

$$\rho_n = 1.0 \times 10^{-7}\,\mathrm{m}$$

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